Scalability & Future Plan

Date	2 July, 2026
Team ID	
Project Name	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current HydraShield – Intelligent Flood Prediction and Early Warning System can be scaled to support larger user bases, increased environmental data, and advanced disaster management features. It also presents the future roadmap for enhancing the system beyond its current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address(High / Medium / Low)
1	Uses historical dataset only without real-time weather integration.	Prediction accuracy depends on historical data availability.	High
2	Single-server Flask deployment without load balancing.	Limited ability to handle a large number of concurrent users.	Medium
3	No automated alert or notification system.	Users must manually check prediction results.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports multiple users on a local Flask server.	Deploy on IBM Cloud, AWS, or Azure with auto-scaling and load balancing to support thousands of concurrent users.
2	Data Integration	Uses a static historical dataset.	Integrate real-time weather APIs and IoT sensor data for live flood prediction.
3	Performance	Prediction is performed synchronously.	Introduce asynchronous processing, model optimization, and caching to reduce response time.

4	Notification System	Prediction results are displayed only on the web application.	Implement SMS, email, and mobile push notifications for real-time flood alerts.

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target timeline**	**Expected Impact**
Phase-2	Integrate real-time weather APIs and IoT sensor data.	Future Enhancement	Improve prediction accuracy with live environmental data.
Phase-3	Develop Android/iOS mobile application and multilingual user interface.	Future Enhancement	Increase accessibility and user reach.
Phase-4	Implement GIS-based flood mapping, advanced AI models, and automated emergency alert systems.	Future Enhancement	Enhance disaster management capabilities and provide more accurate early warning services.